

Cambridge (CIE) IGCSE Additional Maths



Your notes

Trigonometry

Contents

- * Trigonometric Functions
- * The Unit Circle
- * Graphs of Trigonometric Functions
- * Trigonometric Identities
- * Solving Trigonometric Equations
- * Trigonometric Proof



Your notes

Trigonometric Functions

Trigonometric Functions

What are the trigonometric functions?

- The three standard trigonometric functions are: sine, cosine and tangent
- The functions are many-to-one mappings
 - They do not have inverses unless their domain is restricted

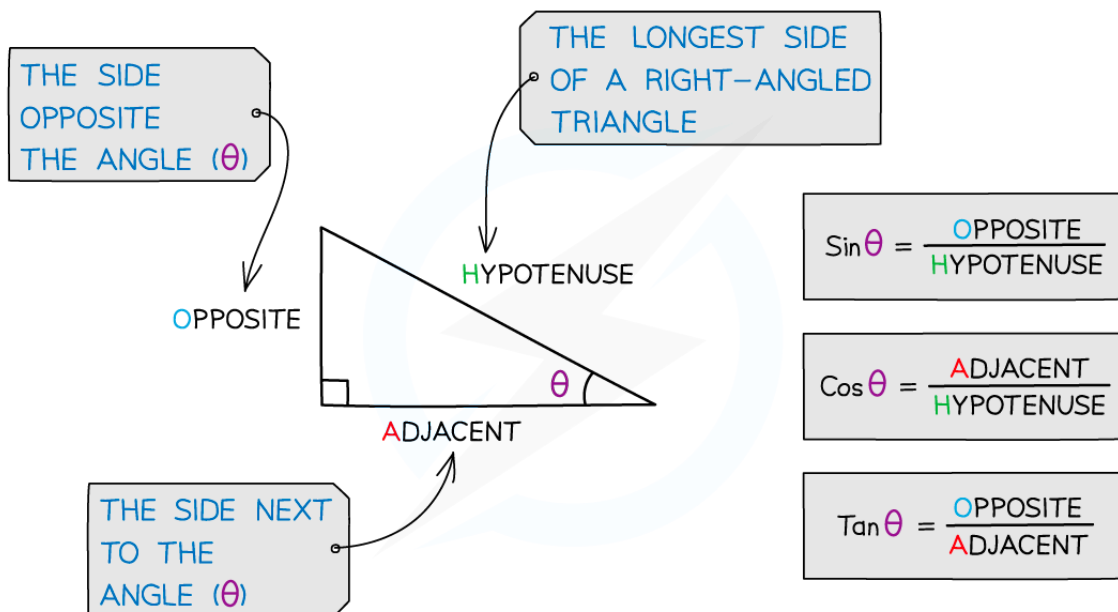
Function	Notation	Domain	Range
Sine	$\sin(x)$	$x \in \mathbb{R}$	$-1 \leq \sin(x) \leq 1$
Cosine	$\cos(x)$	$x \in \mathbb{R}$	$-1 \leq \cos(x) \leq 1$
Tangent	$\tan(x)$	$x \in \mathbb{R}$ $x \neq \pm 90^\circ, \pm 270^\circ, \pm 450^\circ, \dots$ (in degrees) $x \neq \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \dots$ (in radians)	$\tan(x) \in \mathbb{R}$

How can I use the trigonometric functions with right-angled triangles?

- The three trigonometric functions sine, cosine and tangent give the **ratios** of side lengths in right-angled triangles
- For an acute angle in a right-angled triangle:
 - **Sine** of the angle is the length of the side **opposite** the angle divided by the **hypotenuse**
 - **Cosine** of the angle is the length of the side **adjacent** to the angle divided by the **hypotenuse**
 - **Tangent** of the angle is the length of the **opposite** side divided by the length of the **adjacent** side
- You can use the acronym **SOHCAHTOA** to help you remember these



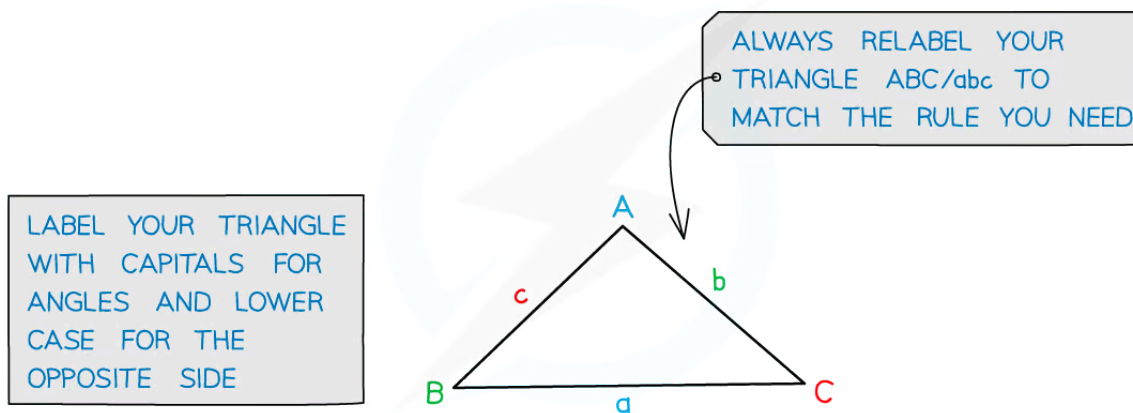
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How can I use the trigonometric functions with non-right-angled triangles?

- You can find missing angles and the lengths of missing sides using the **sine rule** and the **cosine rule**
- You can also use the **area formula**



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SINE RULE

OR

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

TO FIND AN UNKNOWN
SIDE PUT a, b, c ON TOPUSE SINE RULE WHEN YOU
HAVE ANY OPPOSITE PAIRS
OF SIDES AND ANGLESTO FIND AN UNKNOWN ANGLE
PUT $\sin A, \sin B, \sin C$ ON TOPBE CAREFUL! SINE RULE WILL
ALWAYS FIND THE ACUTE ANGLE.
TO FIND THE OBTUSE ANGLE
SUBTRACT FROM 180°

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COSINE RULE

OR

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

TO FIND AN
UNKNOWN SIDE

USE COSINE RULE WHEN
YOU ARE GIVEN AN ANGLE
BETWEEN TWO SIDES
OR ALL THREE SIDES

TO FIND AN
UNKNOWN ANGLE

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AREA OF A TRIANGLE

$$\text{AREA} = \frac{1}{2} ab \sin C$$

YOU COULD BE GIVEN
THE SIDES AND ANGLES
AND ASKED TO FIND AREA

YOU COULD BE TOLD THE AREA AND
ASKED TO FIND A SIDE OR ANGLE

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Reciprocal Trig Functions

What are the reciprocal trig functions?

- There are three reciprocal trig functions that each correspond to either sin, cos or tan
 - **Secant** (sec x)



Your notes

- $\sec x = \frac{1}{\cos x}$

- **Cosecant** (cosec x)

- $\operatorname{cosec} x = \frac{1}{\sin x}$

- **Cotangent** (cot x)

- $\cot x = \frac{1}{\tan x}$

- A good way to remember which function is which is to look at the **third** letter in each of the reciprocal trig functions

- cot **x** is 1 over **t**an **x** etc

- Each of the reciprocal trig functions are **undefined** for certain values of x

- **sec** x is undefined for values of x for which **cos** $x = 0$

- **cosec** x is undefined for values of x for which **sin** $x = 0$

- **cot** x is undefined for values of x for which **tan** $x = 0$

- When $\tan x$ is undefined, $\cot x = 0$

- **Be careful** not to confuse the reciprocal trig functions with the inverse trig functions

- $\sin^{-1} x \neq \frac{1}{\sin x}$



Worked Example

Without the use of a calculator, find the values of

a) $\sec \frac{\pi}{6}$

The third letter of sec is c so it is the reciprocal of cos.

$$\sec\left(\frac{\pi}{6}\right) = \frac{1}{\cos\left(\frac{\pi}{6}\right)}$$



Your notes

Write down the value of $\cos\left(\frac{\pi}{6}\right)$.

$$\sec\left(\frac{\pi}{6}\right) = \frac{1}{\left(\frac{\sqrt{3}}{2}\right)}$$

Simplify.

$$\sec\left(\frac{\pi}{6}\right) = \frac{2}{\sqrt{3}} \text{ or } \frac{2\sqrt{3}}{3}$$

b) $\cot 45^\circ$

The third letter of cot is t so it is the reciprocal of tan.

$$\cot(45^\circ) = \frac{1}{\tan(45^\circ)}$$

Write down the value of $\tan(45^\circ)$.

$$\cot(45^\circ) = \frac{1}{1}$$

$$\cot(45^\circ) = 1$$



Your notes

The Unit Circle

The Unit Circle

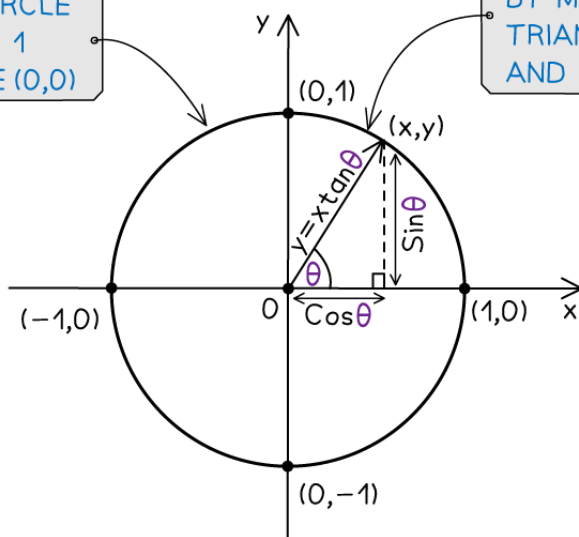
What is the unit circle?

- The unit circle is a circle with radius 1 and centre (0, 0)
- The unit circle can be used to explain how trig functions work with angles that are **not acute**
- Angles are always measured from the positive x-axis and turn:
 - **anticlockwise** for **positive** angles
 - **clockwise** for **negative** angles
- It can be used to calculate trig values as a coordinate point (x, y) on the circle
 - Trig values can be found by making a right triangle with the radius as the hypotenuse
 - Where θ is the angle measured anticlockwise from the positive x-axis
 - The x-axis will always be adjacent to the angle, θ
- SOHCAHTOA can be used to find the values of $\sin\theta$, $\cos\theta$ and $\tan\theta$ easily
- As the radius is 1 unit
 - the **x coordinate** gives the value of **$\cos\theta$**
 - the **y coordinate** gives the value of **$\sin\theta$**
- As the origin is one of the end points - dividing the y coordinate by the x coordinate gives the gradient
 - the **gradient** of the line gives the value of **$\tan\theta$**
- It allows us to calculate sin, cos and tan for angles greater than 90° ($\frac{\pi}{2}$ rad)



Your notes

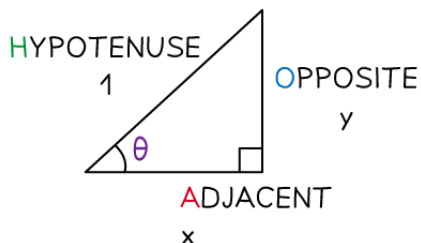
THE UNIT CIRCLE HAS RADIUS 1 AND CENTRE (0,0)



TRIG VALUES CAN BE FOUND BY MAKING A RIGHT ANGLED TRIANGLE WITH THE RADIUS AND x-AXIS

THE ANGLE IS MEASURED ANTI-CLOCKWISE FROM THE x-AXIS IN EITHER RADIANS OR DEGREES

ANY POINT (x,y) ON THE UNIT CIRCLE CAN BE FOUND USING $(\cos \theta, \sin \theta)$



$$\cos \theta = \frac{A}{H} = \frac{x}{1} = x$$

$$\sin \theta = \frac{O}{H} = \frac{y}{1} = y$$

$$\tan \theta = \frac{O}{A} = \frac{y}{x}$$



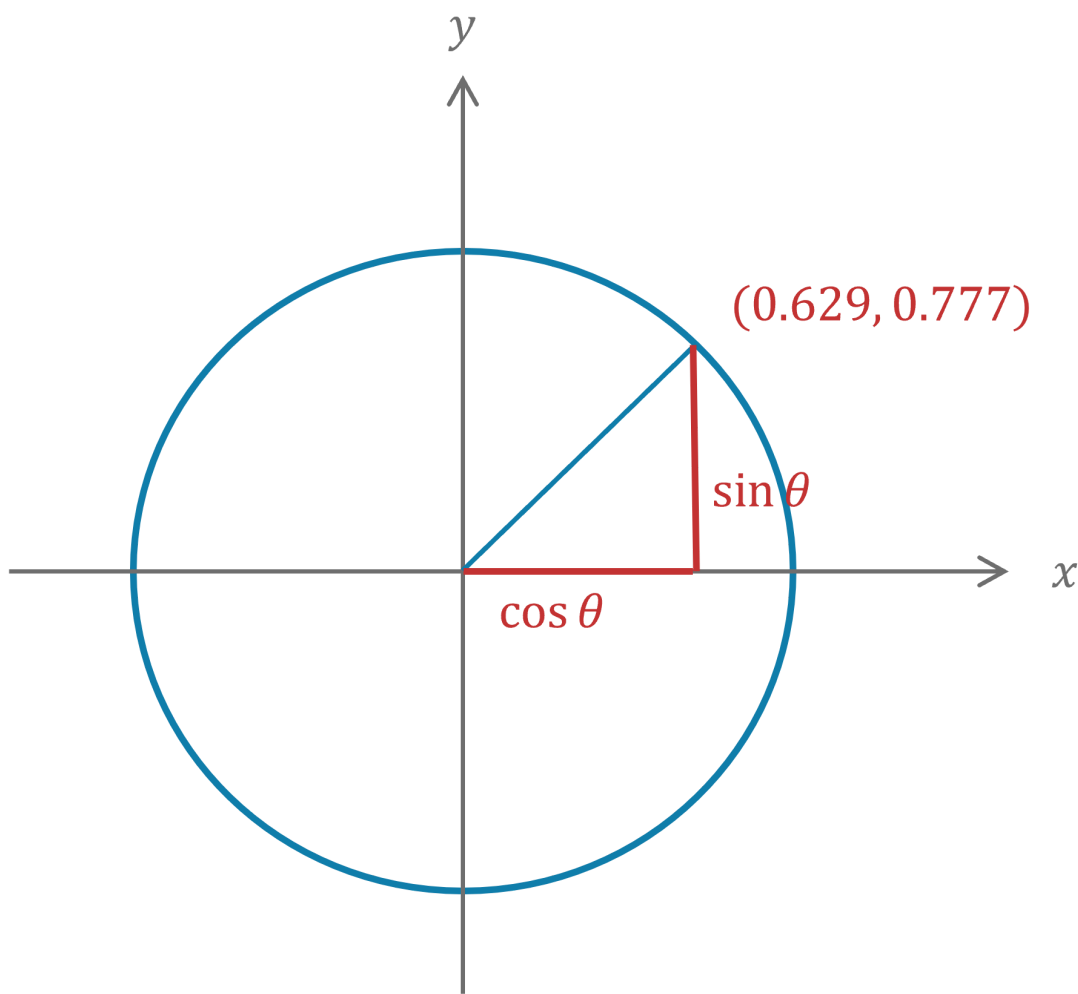
Worked Example

The coordinates of a point on a unit circle, to 3 significant figures, are (0.629, 0.777).

Find θ in radians correct to 2 significant figures..



Your notes



We know that $(x, y) = (\cos \theta, \sin \theta)$ so we can use either ratio to solve for θ .

$$\cos \theta = 0.629$$

$$\theta = \cos^{-1} 0.629$$

$$= 0.89053\dots$$

or

$$\sin \theta = 0.777$$

$$\theta = \sin^{-1} 0.777$$

$$= 0.88988\dots$$

$$\theta = 0.89 \text{ rad (2 s.f.)}$$



Your notes

Using The Unit Circle

What are the properties of the unit circle?

- The unit circle can be split into four **quadrants** at every 90° ($\frac{\pi}{2}$ rad)
 - The first quadrant is for angles between 0 and 90°
 - All three of $\sin\theta$, $\cos\theta$ and $\tan\theta$ are positive in this quadrant
 - The second quadrant is for angles between 90° and 180° ($\frac{\pi}{2}$ rad and π rad)
 - $\sin\theta$ is positive in this quadrant
 - The third quadrant is for angles between 180° and 270° (π rad and $\frac{3\pi}{2}$)
 - $\tan\theta$ is positive in this quadrant
 - The fourth quadrant is for angles between 270° and 360° ($\frac{3\pi}{2}$ rad and 2π)
 - $\cos\theta$ is positive in this quadrant
- Starting from the **fourth** quadrant (on the bottom right) and working anti-clockwise the positive trig functions spell out **CAST**
 - This is why it is often thought of as the **CAST** diagram
 - You may have your own way of remembering this
 - A popular one starting from the first quadrant is **All Students Take Calculus**
- To help picture this better try sketching all three trig graphs on one set of axes and look at which graphs are positive in each 90° section

How is the unit circle used to find secondary solutions?

- Trigonometric functions have **more than one input** to each output
 - For example $\sin 30^\circ = \sin 150^\circ = 0.5$
 - This means that trigonometric equations have **more than one solution**
 - For example both 30° and 150° satisfy the equation $\sin x = 0.5$



Your notes

- The unit circle can be used to find all solutions to trigonometric equations in a given interval
 - Your calculator will only give you the first solution to a problem such as $x = \sin^{-1}(0.5)$
 - This solution is called the **primary value**
 - However, due to the **periodic** nature of the trig functions there could be an infinite number of solutions
 - Further solutions are called the **secondary values**
 - This is why you will be given a **domain** in which your solutions should be found
 - This could either be in degrees or in radians
 - If you see π or some multiple of π then you must work in radians

- The following steps may help you use the unit circle to find **secondary values**

STEP 1: Draw the angle into the first quadrant using the x or y coordinate to help you

- If you are working with **sin** $x = k$, draw the line from the origin to the circumference of the circle at the point where the **y coordinate** is k
- If you are working with **cos** $x = k$, draw the line from the origin to the circumference of the circle at the point where the **x coordinate** is k
- If you are working with **tan** $x = k$, draw the line from the origin to the circumference of the circle such that the **gradient** of the line is k
 - This will give you the angle which should be measured from the **positive x-axis...**
 - ... anticlockwise for a positive angle
 - ... clockwise for a negative angle

STEP 2: Draw the radius in the other quadrant which has the same...

- ... **x-coordinate** if solving **cos** $x = k$
 - This will be the quadrant which is vertical to the original quadrant
- ... **y-coordinate** if solving **sin** $x = k$
 - This will be the quadrant which is horizontal to the original quadrant
- ... **gradient** if solving **tan** $x = k$
 - This will be the quadrant diagonal to the original quadrant

STEP 3: Work out the size of the second angle, measuring from the positive x-axis

- ... anticlockwise for a positive angle

- ... clockwise for a negative angle

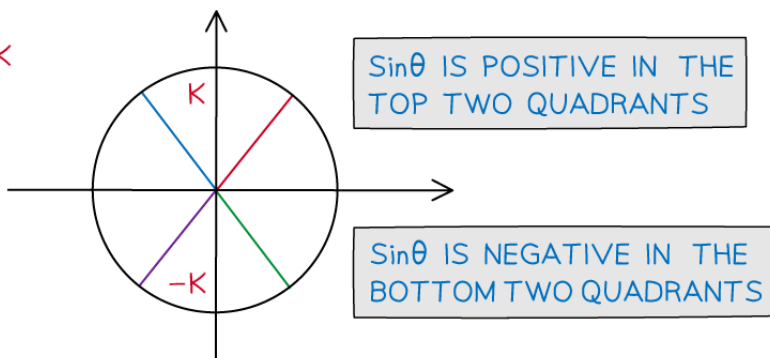
- You should look at the given range of values to decide whether you need the negative or positive angle

STEP 4: Add or subtract either 360° or 2π radians to both values until you have all solutions in the required range

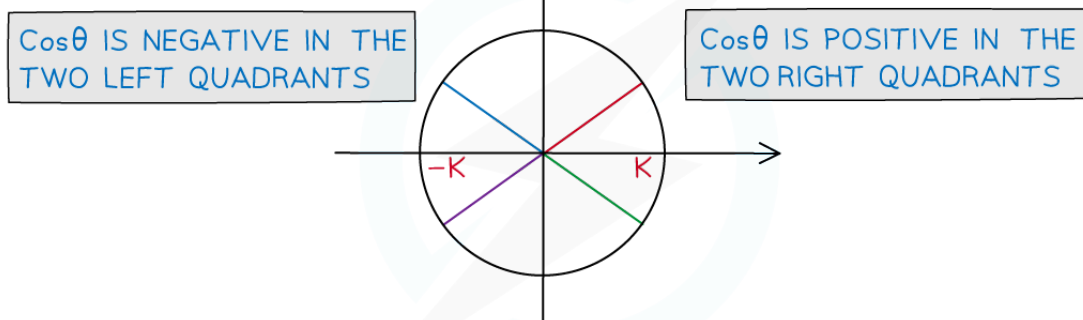


Your notes

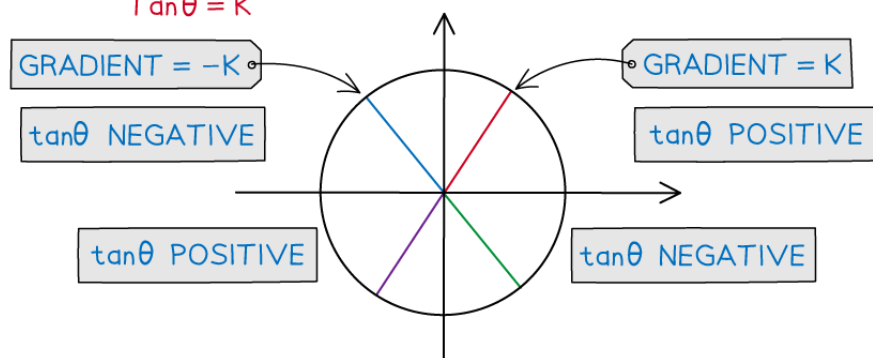
$$\sin \theta = K$$



$$\cos \theta = K$$



$$\tan \theta = K$$



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Examiner Tips and Tricks

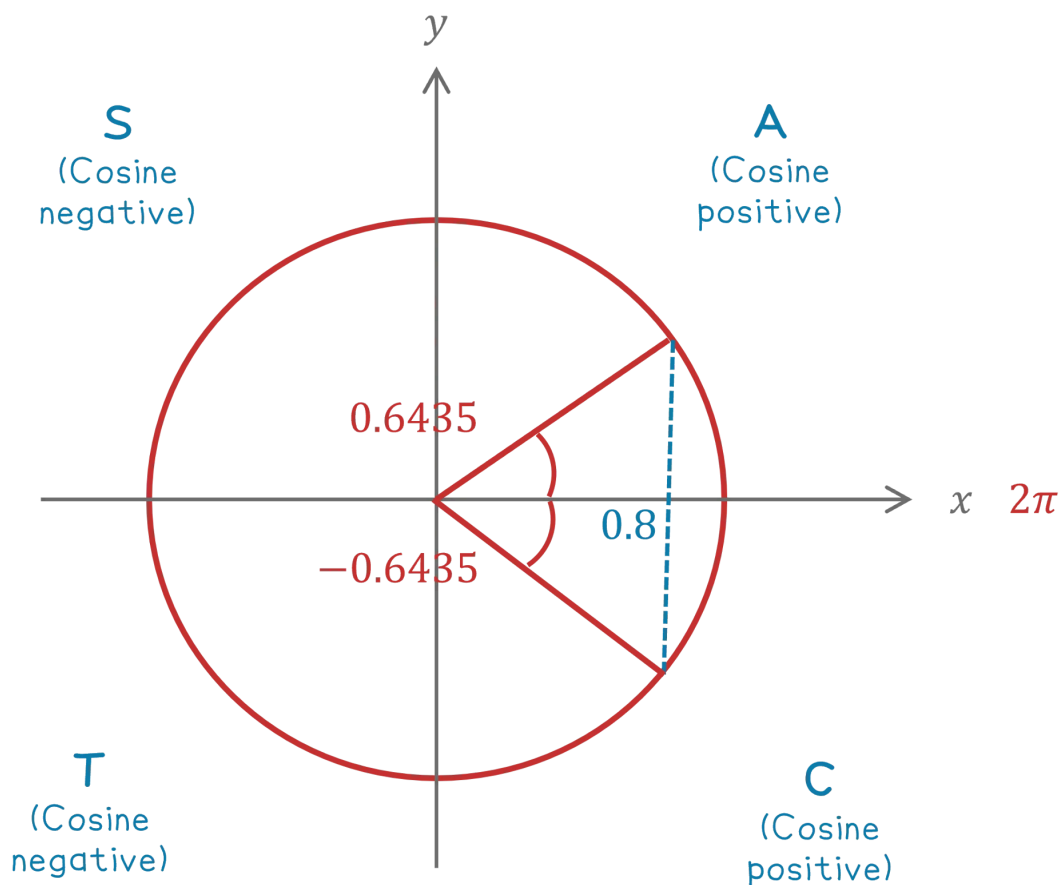
- Being able to sketch out the unit circle and remembering CAST can help you to find all solutions to a problem in an exam question



Worked Example

Given that one solution of $\cos\theta = 0.8$ is $\theta = 0.6435$ radians correct to 4 decimal places, find all other solutions in the range $-2\pi \leq \theta \leq 2\pi$. Give your answers correct to 3 significant figures.

Cosine is positive in the first and fourth quadrants so draw the angle from the horizontal axis in both quadrants.

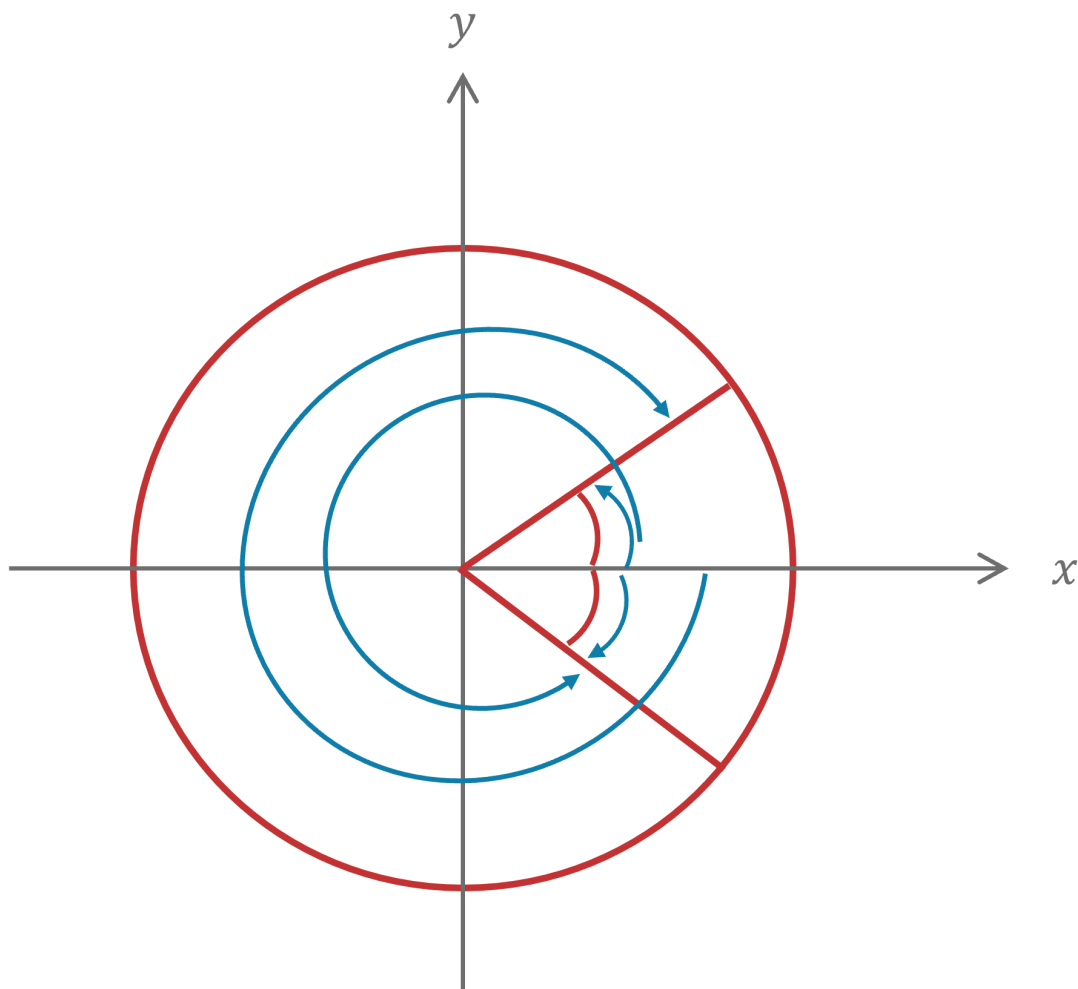




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The primary value is 0.6435 and a secondary value is -0.6435 .

Consider all four possible angles that can be made from the horizontal axis to each of the lines drawn. The range is $-2\pi \leq \theta \leq 2\pi$ so the angles can be in both the positive and negative direction.



The angle in the positive direction to the line in the fourth quadrant is $2\pi - 0.6435 = 5.63968\dots$ The angle in the negative direction to the line in the first quadrant is $-2\pi + 0.6435 = -5.63968\dots$

Solutions are -5.64 , -0.644 , 0.644 , 5.64 rads (3 s.f.)



Your notes

Graphs of Trigonometric Functions

Graphs of Trig Functions

What are the graphs of trigonometric functions?

- The trigonometric functions $\sin$, $\cos$ and $\tan$ all have special periodic graphs
- You'll need to know their properties and how to sketch them for a given domain in either **degrees** or **radians**
- Sketching the trigonometric graphs can help to
 - Solve trigonometric equations and find all solutions
 - Understand transformations of trigonometric functions

What are the properties of the graphs of $\sin x$ and $\cos x$?

- The graphs of $\sin x$ and $\cos x$ are both **periodic**
 - They **repeat every 360° (2π radians)**
 - The angle will always be on the x -axis
 - Either in degrees or radians
- The graphs of $\sin x$ and $\cos x$ are always in the **range $-1 \leq y \leq 1$**
 - **Domain:** $\{x \mid x \in \mathbb{R}\}$
 - **Range:** $\{y \mid -1 \leq y \leq 1\}$
 - The graphs of $\sin x$ and $\cos x$ are identical however one is a **translation** of the other
 - $\sin x$ passes through the origin
 - $\cos x$ passes through $(0, 1)$
- The **amplitude** of the graphs of $\sin x$ and $\cos x$ is 1

What are the properties of the graph of $\tan x$?

- The graph of $\tan x$ is **periodic**
 - It **repeats every 180° (π radians)**



Your notes

- The angle will always be on the x-axis
 - Either in degrees or radians
- The graph of $\tan x$ is **undefined** at the points $\pm 90^\circ, \pm 270^\circ$ etc
 - There are **asymptotes** at these points on the graph
 - In radians this is at the points $\pm \frac{\pi}{2}, \pm \frac{3\pi}{2}$ etc
- The range of the graph of $\tan x$ is
 - Domain: $\left\{ x \mid x \neq \frac{\pi}{2} + k\pi, k \in \mathbb{Z} \right\}$
 - Range: $\{ y \mid y \in \mathbb{R} \}$

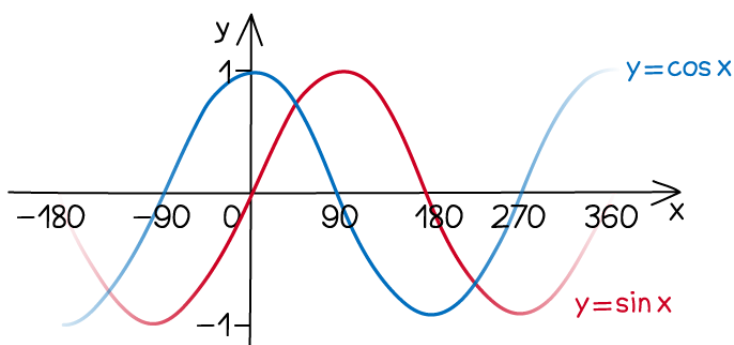


Your notes

$$y = \sin x \quad \text{AND} \quad y = \cos x$$

$\sin x$ AND $\cos x$ ARE ALWAYS
IN THE RANGE -1 TO 1

$\sin x$ PASSES THROUGH THE ORIGIN
 $\cos x$ PASSES THROUGH 1

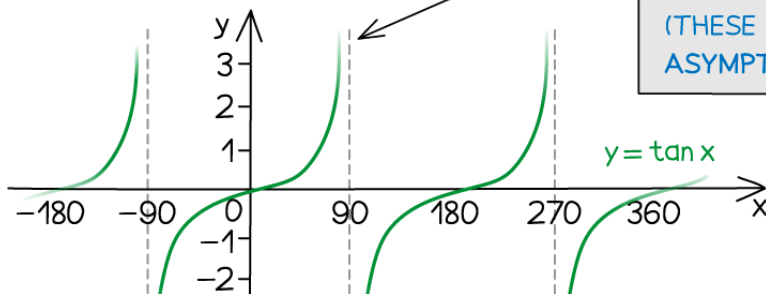


$\sin x$ AND $\cos x$
ARE PERIODIC
REPEATING EVERY 360°

$\sin x$ HAS ROTATIONAL SYMMETRY ABOUT
THE ORIGIN SO $\sin(-x) = -\sin(x)$
 $\cos x$ IS SYMMETRICAL THROUGH THE y -AXIS
SO $\cos(-x) = \cos(x)$

$$y = \tan x$$

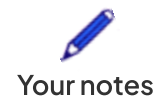
$\tan x$ IS UNDEFINED AT $\pm 90^\circ$,
 $\pm 270^\circ$, $\pm 450^\circ$... MEANING IT
RANGES FROM $-\infty$ TO $+\infty$
(THESE POINTS ARE CALLED
ASYMPTOTES)





$\tan x$ IS PERIODIC
REPEATING EVERY 180°

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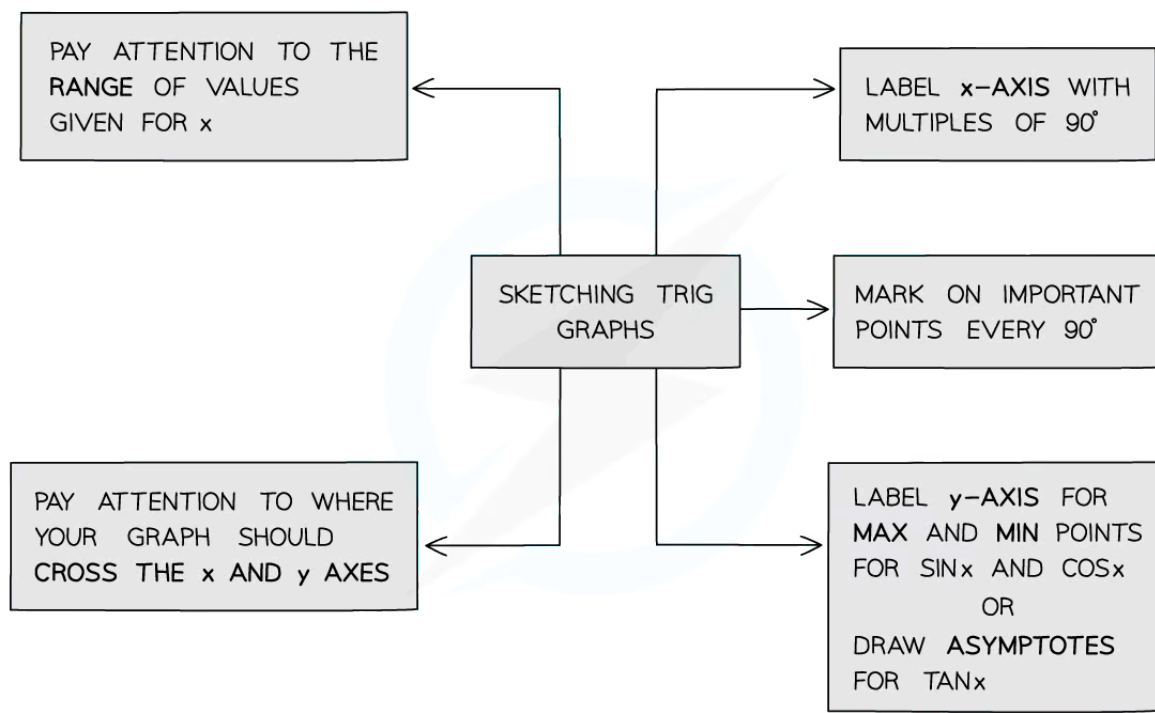


How do I sketch trigonometric graphs?

- You may need to sketch a trigonometric graph so you will need to remember the key features of each one
- The following steps may help you sketch a trigonometric graph
 - STEP 1: Check whether you should be working in degrees or radians
 - You should check the domain given for this
 - If you see π in the given domain then you should work in radians
 - STEP 2: Label the x-axis in multiples of 90°
 - This will be multiples of $\frac{\pi}{2}$ if you are working in radians
 - Make sure you cover the whole domain on the x-axis
 - STEP 3: Label the y-axis
 - The range for the y-axis will be $-1 \leq y \leq 1$ for sin or cos
 - For tan you will not need any specific points on the y-axis
 - STEP 4: Draw the graph
 - Knowing exact values will help with this, such as remembering that $\sin(0) = 0$ and $\cos(0) = 1$
 - Mark the important points on the axis first
 - If you are drawing the graph of $\tan x$ put the asymptotes in first
 - If you are drawing $\sin x$ or $\cos x$ mark in where the maximum and minimum points will be
 - Try to keep the symmetry and rotational symmetry as you sketch, as this will help when using the graph to find solutions



Your notes



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How do I use trigonometric graphs?

- By sketching the graph you can read off all the solutions in a given range (or interval)
- Your calculator will only give you the principal value
- However, due to the **periodic** nature of the trig functions there could be an infinite number of solutions
 - Further solutions are called the **secondary values**
- This is why you will be given a **domain** (interval) in which your solutions should be found
 - This could either be in degrees or in radians
 - If you see π or some multiple of π then you must work in radians
- The following steps will help you use the **trigonometric graphs** to find **secondary values**
 - STEP 1: Sketch the graph for the given function and interval
 - Check whether you should be working in degrees or radians and label the axes with the key values



Your notes

- STEP 2: Draw a horizontal line going through the y-axis at the point you are trying to find the values for
 - For example if you are looking for the solutions to $\sin^{-1}(-0.5)$ then draw the horizontal line going through the y-axis at -0.5
 - The number of times this line cuts the graph is the number of solutions within the given interval
- STEP 3: Find the primary value and mark it on the graph
 - This will either be an exact value and you should know it
 - Or you will be able to use your calculator to find it
- STEP 4: Use the symmetry of the graph to find all the solutions in the interval by adding or subtracting from the key values on the graph
- You should recognise any values/angles associated with **exact values**
- You should be able to spot the pattern of solutions using the **symmetry** and **periodicity** of the graph

STEP 1: SKETCH THE GRAPH FOR THE GIVEN FUNCTION AND RANGE

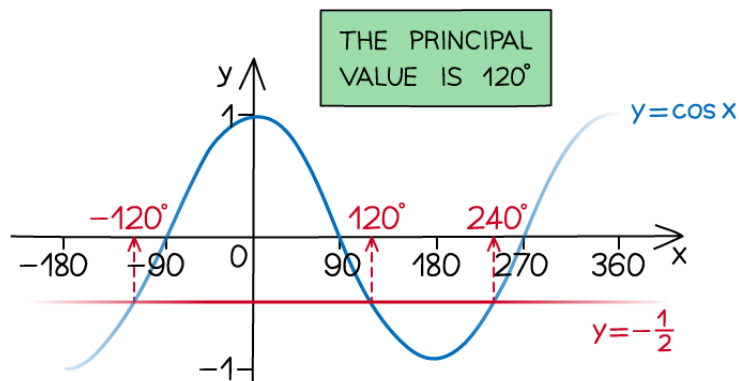
STEP 2: DRAW A HORIZONTAL LINE TO SEE WHERE THE SOLUTIONS SHOULD BE, YOU MAY BE ABLE TO READ DIRECTLY OFF THE GRAPH TO FIND ALL OF THE SOLUTIONS

STEP 3: IF NOT, USE YOUR CALCULATOR TO FIND THE PRINCIPAL VALUE (YOU MAY ALREADY KNOW IT IF IT IS A TRIG EXACT VALUE) AND MARK THIS ON THE GRAPH

STEP 4: USE THE SYMMETRY OF THE GRAPH TO FIND ALL OTHER SOLUTIONS BY ADDING/SUBTRACTING FROM POINTS OF REPETITION WITHIN THE SPECIFIED RANGE

FOR THE SOLUTIONS TO THE EQUATION $\cos x = -\frac{1}{2}$ IN THE RANGE $-180^\circ \leq x \leq 360^\circ$

HERE THE FUNCTION IS $y = \cos x$ AND THE RANGE IS $-180^\circ \leq x \leq 360^\circ$



DRAW THE LINE $y = -\frac{1}{2}$



Your notes

ADD AND SUBTRACT 120° FROM THE POINTS OF REPETITION: 0° AND 360° GIVING SOLUTIONS OF -120° , 120° , AND 240°

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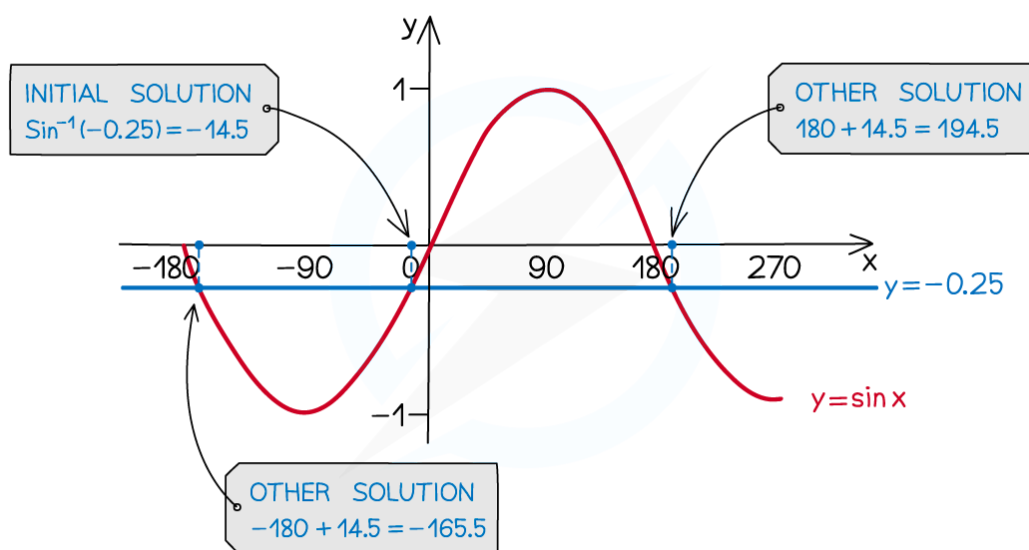
- Always sketch with a pencil, draw a smooth curve and pay attention to the key features of each graph:
 - Where it crosses the x and y axes
 - How often it repeats
 - Whether it is symmetrical
- Remember, when answering exam questions that ask for solutions, a sketch will help ensure you give **all** the appropriate solutions for a given interval



Worked Example



By sketching a graph, find all the solutions to $\sin x = -0.25$ for $-180^\circ \leq x \leq 270^\circ$.



ALL SOLUTIONS ARE:
 $x = -165.5$, $x = -14.5$ AND $x = 194.5$

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Transformations of Trig Functions

What transformations of trigonometric functions do I need to know?

- As with other graphs of functions, trigonometric graphs can be transformed through **translations**, **stretches** and **reflections**
- Translations** can be either horizontal (parallel to the x -axis) or vertical (parallel to the y -axis)
 - For the function $y = \sin(x)$
 - A **vertical** translation of **a units** in the **positive direction** (up) is denoted by $y = \sin(x) + a$



Your notes

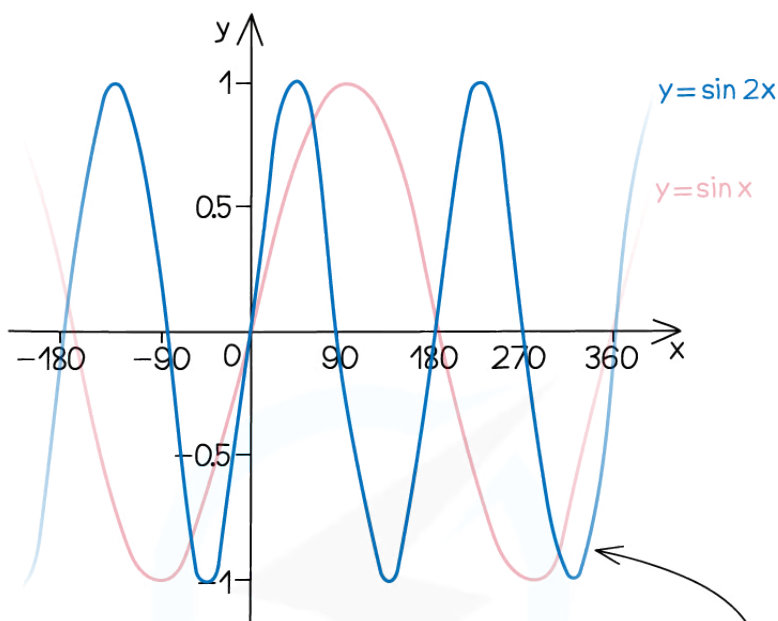
- A **vertical** translation of **a units** in the **negative direction** (down) is denoted by $y = \sin(x) - a$
- A **horizontal** translation in the **positive direction** (right) is denoted by $y = \sin(x - a)$
- A **horizontal** translation in the **negative direction** (left) is denoted by $y = \sin(x + a)$
- **Stretches** can be either horizontal (parallel to the x-axis) or vertical (parallel to the y-axis)
 - For the function $y = \sin(x)$
 - A **vertical** stretch of a **factor a units** is denoted by $y = a \sin(x)$
 - A **horizontal** stretch of a **factor a units** is denoted by $y = \sin\left(\frac{x}{a}\right)$
- **Reflections** can be either across the x-axis or across the y-axis
 - For the function $y = \sin(x)$
 - A **reflection** across the x-axis is denoted by $y = -\sin(x)$



Your notes

$y = \sin nx$ GIVES A
HORIZONTAL STRETCH
OR SQUASH

THE GRAPH OF $y = \sin x$
SQUASHED HORIZONTALLY
BY A FACTOR OF 2
MEANING IT REPEATS
ITSELF TWICE EVERY 360°



STEP 1: SKETCH THE ORIGINAL $\sin$, $\cos$ OR $\tan$ FUNCTION
STEP 2: PICK KEY CO-ORDINATE POINTS TO PLOT
THE NEW GRAPH
STEP 3: SKETCH IN THE NEW TRANSFORMED GRAPH

FOR A HORIZONTAL STRETCH (OR SQUASH)

- IF $n > 1$, THE GRAPH IS SQUASHED HORIZONTALLY BY A SCALE FACTOR OF n
- IF $0 < n < 1$, THE GRAPH IS STRETCHED BY THAT SCALE FACTOR
- IF $n < 0$, THE GRAPH IS ALSO REFLECTED IN THE y -AXIS
- REMEMBER THE y COORDINATES STAY THE SAME;

x COORDINATES ARE MULTIPLIED BY $\frac{1}{n}$

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Your notes

What combined transformations are there?

- **Stretches** in the horizontal and vertical direction are often combined
- The functions **$a \sin(bx)$** and **$a \cos(bx)$** have the following properties:
 - The **amplitude** of the graph is $|a|$
 - The **period** of the graph is $\frac{360}{b}^\circ$ (or $\frac{2\pi}{b}$ rad)
 - In this course, a will always be a positive integer and b will be a simple fraction or integer
- **Translations** in both directions could also be combined with the stretches
- The functions **$a \sin bx + c$** and **$a \cos bx + c$** have the following properties:
 - The **amplitude** of the graph is $|a|$
 - The **period** of the graph is $\frac{360}{b}^\circ$ (or $\frac{2\pi}{b}$)
 - The translation in the vertical direction is c
 - c represents the **principal axis** (the line that the function fluctuates about)
 - It helps to start by sketching the principle axis
- The function **$a \tan bx + c$** has the following properties:
 - The **amplitude** of the graph does not exist
 - The **period** of the graph is $\frac{180}{b}^\circ$ (or $\frac{\pi}{b}$)
 - The translation in the vertical direction (principal axis) is c
 - Finding and drawing the asymptotes first can help to sketch these graphs

How do I sketch transformations of trigonometric functions?

- Sketch the graph of the original function first
- Carry out each transformation separately



Your notes

- The **order** in which you carry out the transformations is important
- Given the form $y = a \sin bx + c$ carry out any **stretches** first, **translations** next and **reflections** last
- Use a very light pencil to mark where the graph has moved for each transformation
- It is a good idea to mark in the principal axis the lines corresponding to the maximum and minimum points first
 - The **principal axis** will be the line $y = c$
 - The **maximum points** will be on the line $y = c + a$
 - The **minimum points** will be on the line $y = c - a$
- Sketch in the new transformed graph
- Check it is correct by looking at some key points from the **exact values**



Examiner Tips and Tricks

- Always sketch with a pencil and draw a smooth curve
- When you sketch the transformation of a graph, be sure to indicate the new coordinates of any points that are marked on the original graph
 - For any graph involving tan the asymptotes must be clearly labelled
- Try to indicate the coordinates of points where the new graph intersects the axes



Worked Example

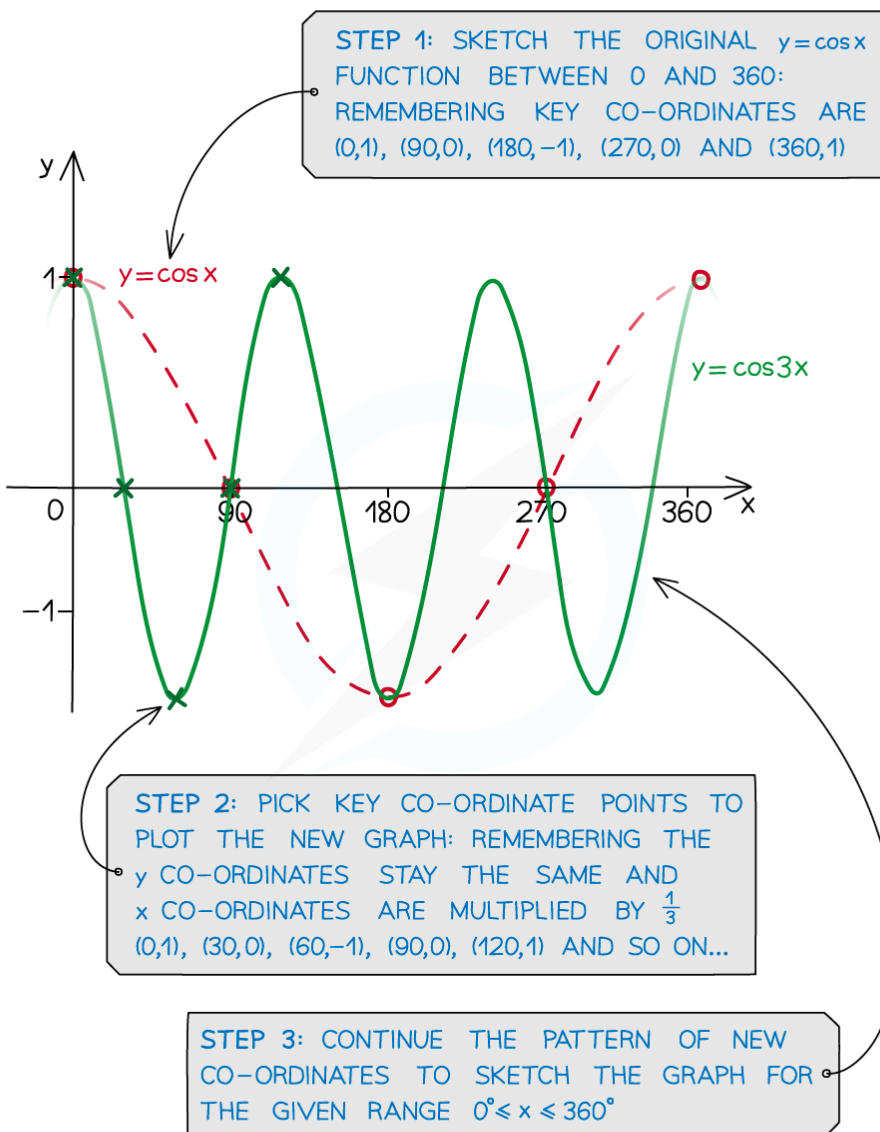


Your notes



- a) Sketch the graph of $y = \cos 3x$, for $0^\circ \leq x \leq 360^\circ$.
b) Write down all the values of x where $\cos 3x = 0$, for $0^\circ \leq x \leq 360^\circ$.

a)



b) (30,0), (90,0), (150,0), (210,0), (270,0), (330,0)

LIST OUT THE CO-ORDINATES WHERE
 $y = \cos 3x$ CROSSES THE x-AXIS, AGAIN
YOU SHOULD BE ABLE TO SEE A
PATTERN (+60 EACH TIME)

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Your notes



Your notes

Trigonometric Identities

Simple Trig Identities

What is a trigonometric identity?

- Trigonometric identities are statements that are true for all values of x or θ
- They are used to help simplify trigonometric equations before solving them
- Sometimes you may see identities written with the symbol $\equiv$
 - This means 'identical to'

What trigonometric identities do I need to know?

- The two trigonometric identities you must know are
 - $\tan \theta = \frac{\sin \theta}{\cos \theta}$
 - This is the identity for $\tan \theta$
 - This formula does **not** appear in the list of formulae
 - $\sin^2 \theta + \cos^2 \theta = 1$
 - This is the Pythagorean identity
 - Note that the notation $\sin^2 \theta$ is the same as $(\sin \theta)^2$
 - This formula appears in the list of formulae
- Rearranging the second identity often makes it easier to work with
 - $\sin^2 \theta = 1 - \cos^2 \theta$
 - $\cos^2 \theta = 1 - \sin^2 \theta$

Where do the trigonometric identities come from?

- You do not need to know the proof for these identities but it is a good idea to know where they come from
- From SOHCAHTOA we know that



Your notes

- $\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{O}{H}$
- $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{A}{H}$
- $\tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{O}{A}$
- The identity for $\tan \theta$ can be seen by dividing $\sin \theta$ by $\cos \theta$?
 - $\frac{\sin \theta}{\cos \theta} = \frac{\frac{O}{H}}{\frac{A}{H}} = \frac{O}{A} = \tan \theta$
- This can also be seen from the unit circle by considering a right-triangle with a hypotenuse of 1
 - $\tan \theta = \frac{O}{A} = \frac{\sin \theta}{\cos \theta}$
- The Pythagorean identity can be seen by considering a right-triangle with a hypotenuse of 1
 - Then $(\text{opposite})^2 + (\text{adjacent})^2 = 1$
 - Therefore $\sin^2 \theta + \cos^2 \theta = 1$
- Considering the equation of the unit circle also shows the Pythagorean identity
 - The equation of the unit circle is $x^2 + y^2 = 1$
 - The coordinates on the unit circle are $(\cos \theta, \sin \theta)$
 - Therefore the equation of the unit circle could be written $\cos^2 \theta + \sin^2 \theta = 1$

How are the trigonometric identities used?

- Most commonly trigonometric identities are used to change an equation into a form that allows it to be solved
- They can also be used to prove further identities



Examiner Tips and Tricks

- If you are asked to show that one thing is identical ($\equiv$) to another, look at what parts are missing



Worked Example

Show that the equation $2\sin^2 x - \cos x = 0$ can be written in the form $a\cos^2 x + b\cos x + c = 0$, where a , b and c are integers to be found.

Substitute $\sin^2 x = 1 - \cos^2 x$ into the equation to form a quadratic in terms of $\cos x$.

$$2(1 - \cos^2 x) - \cos x = 0$$

Expand the bracket.

$$2 - 2\cos^2 x - \cos x = 0$$

Rewrite in the required form.

$$2\cos^2 x + \cos x - 2 = 0$$

or

$$-2\cos^2 x - \cos x + 2 = 0$$

Further Trig Identities

What are the identities linking $\tan$, $\sec$, $\cot$, and cosec ?

- Aside from the Pythagorean identity $\sin^2 x + \cos^2 x = 1$ there are two further Pythagorean identities you will need to learn
 - $1 + \tan^2 \theta = \sec^2 \theta$
 - $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$
 - Both can be found in the list of formulae
- Both of these identities can be derived from $\sin^2 x + \cos^2 x = 1$
 - To derive the identity for $\sec^2 x$ divide $\sin^2 x + \cos^2 x = 1$ by $\cos^2 x$
 - To derive the identity for $\operatorname{cosec}^2 x$ divide $\sin^2 x + \cos^2 x = 1$ by $\sin^2 x$

$$\sin^2 x + \cos^2 x = 1$$

$\div \cos^2 x$

$$\frac{\sin^2 x}{\cos^2 x} + \frac{\cos^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}$$

$$\tan^2 x + 1 = \sec^2 x$$

$\div \sin^2 x$

$$\frac{\sin^2 x}{\sin^2 x} + \frac{\cos^2 x}{\sin^2 x} = \frac{1}{\sin^2 x}$$

$$1 + \cot^2 x = \operatorname{cosec}^2 x$$

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How do I prove new trigonometric identities?

- You can use trigonometric identities you already know to prove new identities
- To prove an identity start on one side and proceed step by step until you get to the other side
 - It is more common to start on the left hand side but you can start a proof from either end
 - Occasionally it is easier to show that one side subtracted from the other is zero
 - You should not work on both sides simultaneously
- Look for anything that could be a part of one of the above identities on either side
 - For example if you see $\sin^2 \theta$ you can replace it with $1 - \cos^2 \theta$
- Look for ways of reducing the number of different trigonometric functions there are within the identity
 - For example if the identity contains $\tan \theta$, $\cot \theta$ and $\operatorname{cosec} \theta$ you could try
 - Using the identities $\tan \theta = 1/\cot \theta$ and $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$ to write it all in terms of $\cot \theta$
 - Or rewriting it all in terms of $\sin \theta$ and $\cos \theta$ and simplifying
- Often you may need to trial a few different methods before finding the correct one
- Always keep an eye on the 'target' expression – this can help suggest what identities to use



Examiner Tips and Tricks

- Writing down all the identities you know can be a good way to spot how to get started on a question



Your notes



Worked Example

Solve the equation $9 \sec^2 \theta - 11 = 3 \tan \theta$ in the interval $0 \leq \theta \leq 2\pi$. Give your answers to three decimal places.

The squared term can be rewritten using the identity $\sec^2 \theta = 1 + \tan^2 \theta$. Substitute this into the equation.

$$9(1 + \tan^2 \theta) - 11 = 3 \tan \theta$$

Rearrange to form a quadratic.

$$9 + 9 \tan^2 \theta - 11 = 3 \tan \theta$$

$$9 \tan^2 \theta - 3 \tan \theta - 2 = 0$$

Treat as a quadratic using $x = \tan \theta$.

$$9x^2 - 3x - 2 = 0$$

$$(3x - 2)(3x + 1) = 0$$

$$x = \frac{2}{3}, x = -\frac{1}{3}$$

Substitute $x = \tan \theta$ back in and find the principal value for each equation.

$$\tan \theta = \frac{2}{3} \rightarrow \tan^{-1}\left(\frac{2}{3}\right) = 0.5880...$$

$$\tan \theta = -\frac{1}{3} \rightarrow \tan^{-1}\left(-\frac{1}{3}\right) = -0.3217...$$

Find the other solutions in the interval by adding multiples of π .

$$\pi + 0.5880... = 3.7295...$$

$$\pi + (-0.3217...) = 2.8198...$$

$$2\pi + (-0.3217...) = 5.9614...$$

Ignore the solution that is outside the interval $(-0.3217\dots)$. Round the other four to three decimal places.

$$\theta = 0.588, 2.820, 3.730, 5.961$$



Your notes



Your notes

Solving Trigonometric Equations

Linear Trig Equations

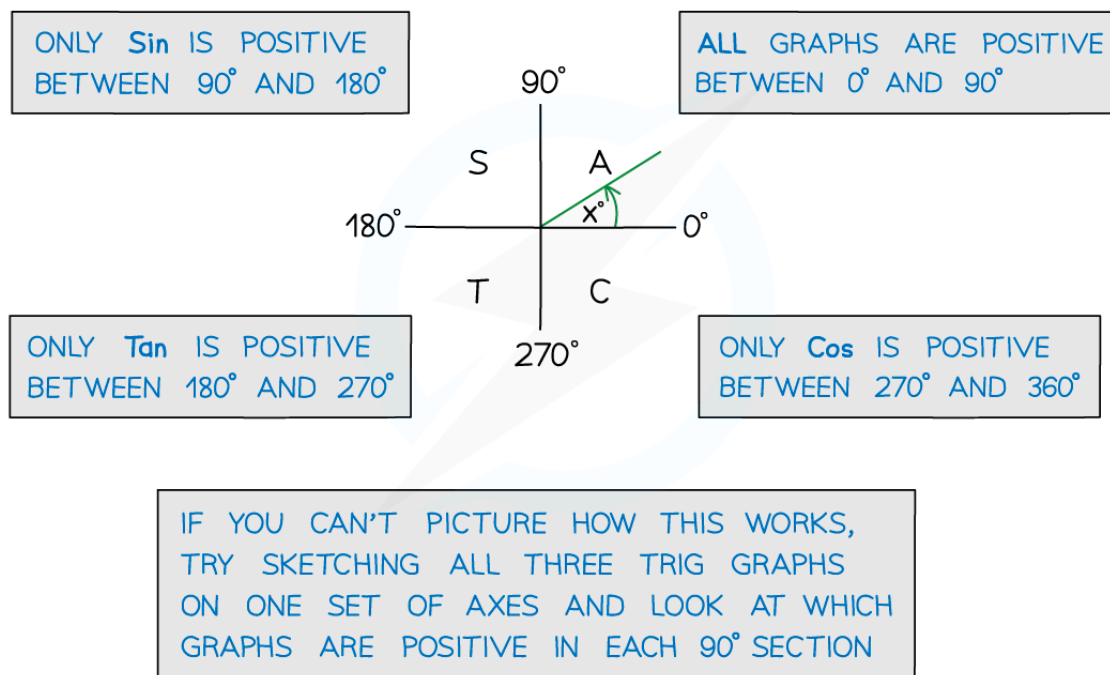
How are trigonometric equations solved?

- Trigonometric equations can have an infinite number of solutions
 - For an equation in $\sin$ or $\cos$ you can add 360° or 2π to each solution to find more solutions
 - For an equation in $\tan$ you can add 180° or π to each solution
- When solving a trigonometric equation you will be given a range of values within which you should find all the values
- Solving the equation normally and using the inverse function on your calculator or your knowledge of **exact values** will give you the **primary value**
- The **secondary values** can be found with the help of:
 - The **unit circle**
 - By using the **CAST** diagram which shows where each function has positive solutions
 - The **graphs of trigonometric functions**
 - By **sketching the graph** (see Graphs of Trigonometric Functions) you can read off all the solutions in a given range (or interval)
- By using **trigonometric identities** you can simplify harder equations
- You may be asked to use **degrees or radians** to solve trigonometric equations
 - Make sure your calculator is in the **correct mode**
 - Remember common angles
 - 90° is $\frac{1}{2}\pi$ radians
 - 180° is π radians
 - 270° is $\frac{3\pi}{2}$ radians
 - 360° is 2π radians



Your notes

THE CAST DIAGRAM



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How are trigonometric equations of the form $\sin x = k$ solved?

- It is a good idea to sketch the graph of the trigonometric function first
 - Use the given range of values as the domain for your graph
 - The intersections of the graph of the function and the line $y = k$ will show you
 - The location of the solutions
 - The number of solutions
 - You will be able to use the symmetry properties of the graph to find all secondary values within the given range of values
- The method for finding secondary values are:
 - For the equation $\sin x = k$ the primary value is $x_1 = \sin^{-1} k$



Your notes

- A secondary value is $x_2 = 180^\circ - \sin^{-1} k$
- Then all values within the range can be found using $x_1 \pm 360n$ and $x_2 \pm 360n$ where $n \in \mathbb{N}$
- For the equation $\cos x = k$ the primary value is $x_1 = \cos^{-1} k$
- A secondary value is $x_2 = -\cos^{-1} k$
- Then all values within the range can be found using $x_1 \pm 360n$ and $x_2 \pm 360n$ where $n \in \mathbb{N}$
- For the equation $\tan x = k$ the primary value is $x = \tan^{-1} k$
- All secondary values within the range can be found using $x \pm 180n$ where $n \in \mathbb{N}$

How do I use the CAST diagram?

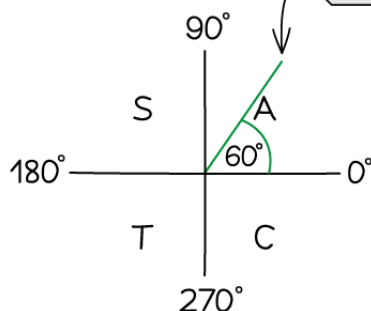


Your notes

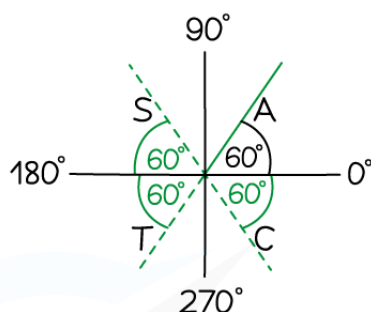
e.g. FIND ALL THE VALUES OF x IN THE INTERVAL
 $-360^\circ \leq x \leq 720^\circ$ WHERE $\cos x = \frac{1}{2}$

PRINCIPAL VALUE
 $\cos^{-1}\left(\frac{1}{2}\right) = 60^\circ$

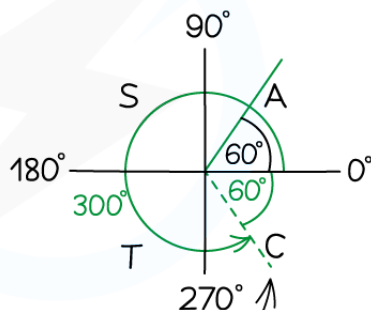
STEP 1: CALCULATE THE PRINCIPLE VALUE AND DRAW IT INTO THE CAST DIAGRAM STARTING FROM 0°



STEP 2: CONTINUE THE LINE ACROSS THE CAST DIAGRAM MAKING THE SAME ANGLE IN ALL 4 QUADRANTS



STEP 3: FOCUSING ON THE LINES IN THE VALID QUADRANTS WORK OUT ALL POSITIVE SOLUTIONS FOR $0^\circ \leq x \leq 360^\circ$



SOLUTIONS FOR
 $\cos x = \frac{1}{2}$ ARE POSITIVE
 IN THE 1st AND 4th
 QUADRANTS ONLY
 $x = 60^\circ, 300^\circ$

STEP 4: CHECK YOUR QUESTION FOR THE GIVEN INTERVAL, IF THIS IS BEYOND $0^\circ \leq x \leq 360^\circ$

FIND ALL SOLUTIONS

YOU MUST ADD AND SUBTRACT
FROM 360° TO FIND ALL SOLUTIONS

BETWEEN $-360^\circ \leq x \leq 720^\circ$

$$x = 60$$

$$60 + 360 = 420$$

$$60 - 360 = -300$$

$$x = 300$$

$$300 + 360 = 660$$

$$300 - 360 = -60$$

$$x = -300^\circ, -60^\circ, 60^\circ, 300^\circ, 420^\circ, 660^\circ$$

ADDING OR SUBTRACTING
ANOTHER 360° FROM ANY
OF THESE SOLUTIONS GIVES
AN ANSWER THAT IS TOO
BIG OR TOO SMALL

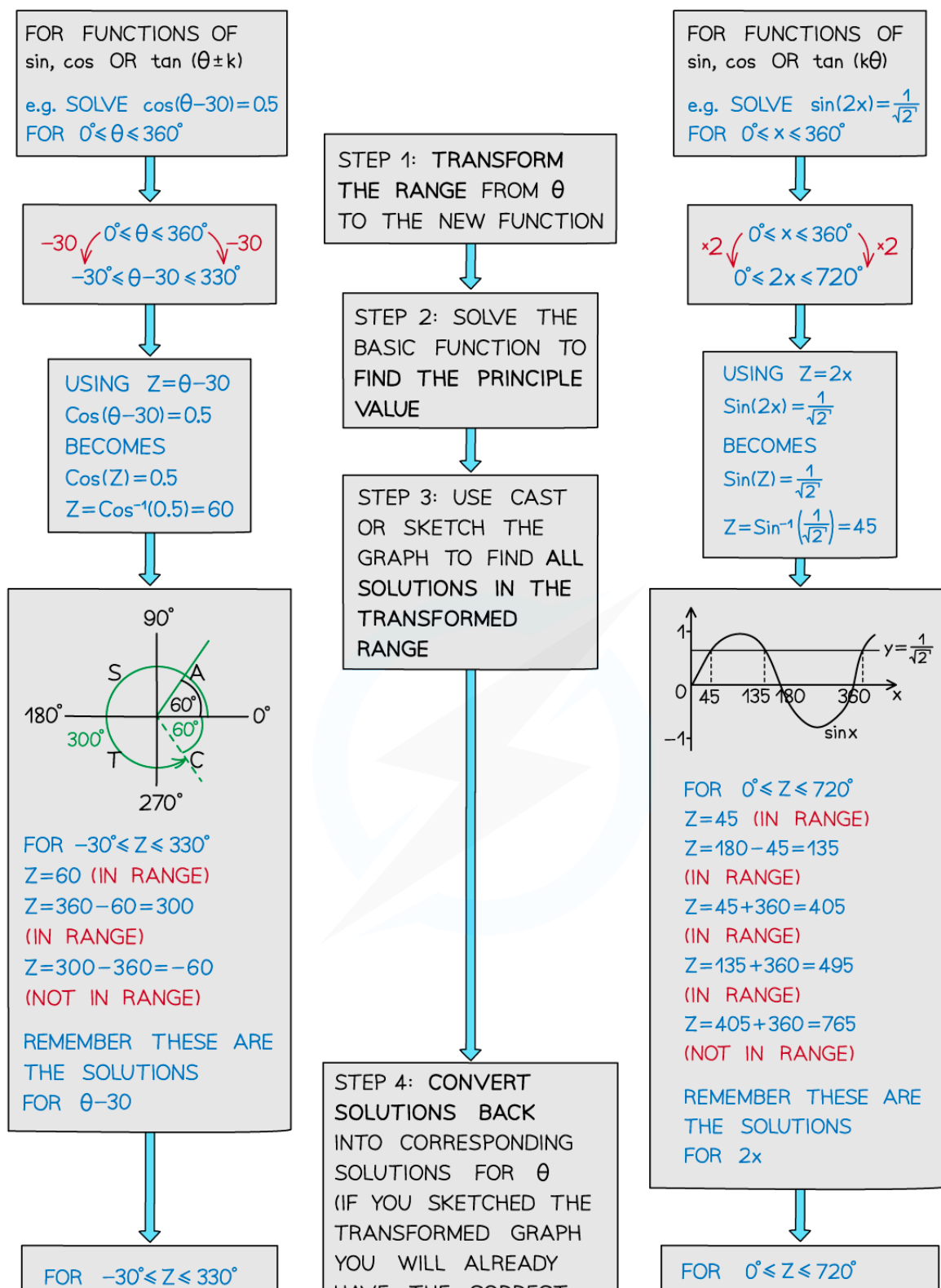
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Your notes

What about more complicated trig equations?

- Some trig equations could involve a **function** of x or θ (see Transformations of Trigonometric Functions)
- Trigonometric equations in the form $\sin(ax + b)$ can be solved in more than one way
- The easiest method is to consider the transformation of the angle as a substitution
 - For example let $u = ax + b$
- Transform the given interval for the solutions in the same way as the angle
 - For example if the given interval is $0^\circ \leq x \leq 360^\circ$ the new interval will be
 - $(a(0^\circ) + b) \leq u \leq (a(360^\circ) + b)$
- Solve the function to find the primary value for u
- Use either the unit circle or sketch the graph to find all the other solutions in the range for u
- Undo the substitution to convert all of the solutions back into the corresponding solutions for x
- Another method would be to sketch the transformation of the function
 - If you use this method then you will not need to use a substitution for the range of values



(WHERE $Z = \theta - 30$)
 $Z = 60$ AND $Z = 300$

$\swarrow +30$ $+30 \searrow$

$\theta = 90^\circ$ AND $\theta = 330^\circ$
 WHICH ARE IN THE
 RANGE $0^\circ \leq \theta \leq 360^\circ$

HAVE THE CORRECT
 SOLUTIONS, JUST CHECK
 THEY ARE IN THE
 CORRECT RANGE)

(WHERE $Z = 2x$)
 $Z = 45, 135, 405, 495$

$\div 2 \downarrow$

$x = 22.5, 67.5, 202.5, 247.5$
 WHICH ARE IN THE
 RANGE $0^\circ \leq x \leq 360^\circ$

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Your notes

What about equations using sec, cosec, and cot?

- Equations in the form $\sec x = k$ (or $\operatorname{cosec} x = k$ or $\cot x = k$) are not solvable using your calculator, as it does not have an inverse function for these
- Use the definitions of these functions to rewrite the equation in terms of $\sin x$, $\cos x$, or $\tan x$
 - and then solve as normal, remembering to find all the solutions within the given domain of x
- It is often useful to swap sec, cosec, and cot for their definitions as one of the first steps when solving an equation



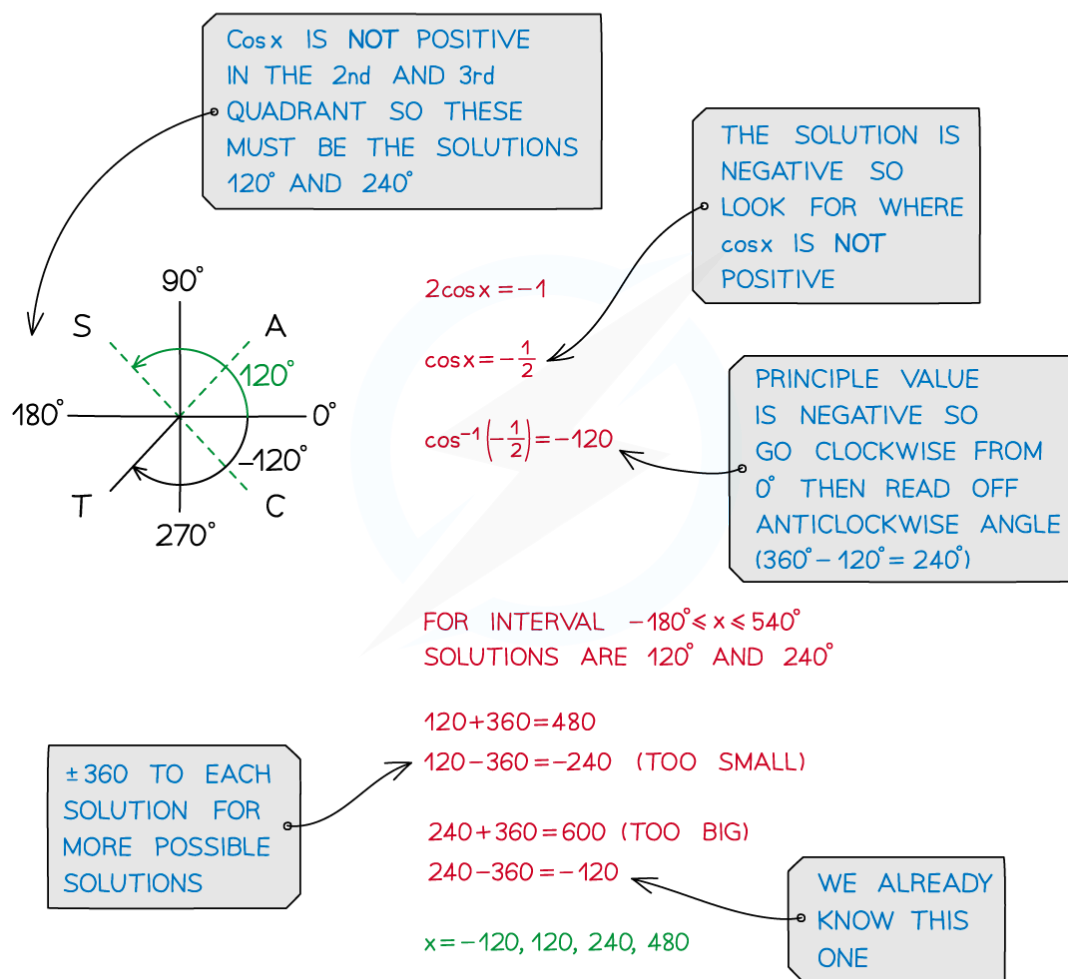
Examiner Tips and Tricks

- Your calculator will only give you the principal value and you need to find all other solutions for the given interval
- Also, remember the CAST diagram only gives you some solutions, so again you may need to find more depending on the given interval
- It is entirely up to you how you solve a trig equation, but some ways are more helpful than others depending on the type of equation you are trying to solve



Worked Example

? Solve the equation $2\cos x = -1$, in the range $-180^\circ \leq x \leq 540^\circ$.



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Quadratic Trig Equations

How are quadratic trigonometric equations solved?



Your notes

- A quadratic trigonometric equation is one that includes either $\sin^2 \theta$, $\cos^2 \theta$ or $\tan^2 \theta$
- Often the **identity** $\sin^2 \theta + \cos^2 \theta = 1$ can be used to rearrange the equation into a form that is possible to solve
 - If the equation involves both sine and cosine then the **Pythagorean identity** should be used to write the equation in terms of just one of these functions
 - If the equation involves a mixture of regular trig functions (sin, cos, and tan) as well as reciprocal functions (sec, cosec, and cot) it may be useful to:
 - rewrite the reciprocal trig functions using their definitions,
 - use the identities $1 + \tan^2 \theta = \sec^2 \theta$ and $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$
- Solve the **quadratic equation** using the quadratic equation or factorisation
 - This can be made easier by changing the function to a single letter
 - Such as changing $2\cos^2 \theta - 3\cos \theta - 1 = 0$ to $2c^2 - 3c - 1 = 0$
- A quadratic can give up to two solutions
 - You must consider both solutions to see whether a real value exists
 - Remember that solutions for $\sin \theta = k$ and $\cos \theta = k$ only exist for $-1 \leq k \leq 1$
 - Solutions for $\tan \theta = k$ exist for all values of k
- Find all solutions within the given interval
 - There will often be more than two solutions for one quadratic equation
 - The best way to check the number of solutions is to sketch the graph of the function
- You may be asked to use **degrees or radians** to solve trigonometric equations
 - Make sure your calculator is in the **correct mode**
 - Remember common angles
 - 90° is $\frac{1}{2}\pi$ radians
 - 180° is π radians
 - 270° is $\frac{3\pi}{2}$ radians
 - 360° is 2π radians

SOLVING QUADRATIC TRIG EQUATIONS

e.g. SOLVE $2\sin^2\theta = -3\cos\theta$
FOR THE INTERVAL
 $0^\circ \leq x \leq 360^\circ$

IF NEEDED USE TRIG IDENTITIES TO REWRITE EQUATIONS SO EVERYTHING IS IN SIMILAR TERMS

USING $\sin^2\theta + \cos^2\theta = 1$
 $2\sin^2\theta = -3\cos\theta$
CAN BE WRITTEN AS
 $2\cos^2\theta - 3\cos\theta - 2 = 0$

WHEN FACTORISING A QUADRATIC TRIG EQUATION, IT MIGHT HELP TO REWRITE IT USING JUST ONE LETTER

$2\cos^2\theta - 3\cos\theta - 2 = 0$
REPLACING $\cos\theta$ WITH JUST C
 $2C^2 - 3C - 2 = 0$
FACTORIZE TO GIVE
 $(2C+1)(C-2) = 0$
THEN REWRITE WITH $\cos\theta$
 $(2\cos\theta + 1)(\cos\theta - 2) = 0$

ALWAYS CHECK WHETHER SOLUTIONS EXIST.
SOLUTIONS TO $\sin x = k$
AND $\cos x = k$ ONLY EXIST WHERE $-1 \leq k \leq 1$.
WHEREAS SOLUTIONS TO $\tan x = k$ EXIST FOR ALL VALUES OF k

$(2\cos\theta + 1)(\cos\theta - 2) = 0$
TO SOLVE
 $2\cos\theta + 1 = 0$ AND $\cos\theta - 2 = 0$
 $\cos\theta = -\frac{1}{2}$ AND $\cos\theta = 2$
NO SOLUTIONS EXIST FOR $\cos\theta = 2$
SO SOLUTIONS CAN ONLY COME FROM $\cos\theta = -\frac{1}{2}$

IF YOU ARE GIVEN AN INTERVAL YOU MUST REMEMBER TO FIND ALL THE POSSIBLE SOLUTIONS WITHIN THE INTERVAL. BE CAREFUL IF YOUR INITIAL SOLUTION IS OUTSIDE THE INTERVAL
e.g. NEGATIVE SOLUTION FOR INTERVAL OF $0^\circ \leq x \leq 360^\circ$

SOLUTIONS FROM $\cos\theta = -\frac{1}{2}$
 $\cos^{-1}\left(-\frac{1}{2}\right) = 120$
FINDING OTHER SOLUTIONS
 $360 - 120 = 240$
 $\theta = 120$ AND 240



Examiner Tips and Tricks

- Sketch the appropriate sin, cos, tan graph to ensure you find ALL solutions **within** the given interval, and be super careful if you get a negative solution but have a positive interval
- For example, for an equation, in the interval $0^\circ \leq x \leq 360^\circ$, with solution $\sin x = -\frac{1}{4}$ then $\sin^{-1}(-\frac{1}{4}) = -14.5$ (to 1d.p.), which is not between 0 and 360
 - by sketching the graph you'll be able to spot the two solutions will be $180 + 14.5$ and $360 - 14.5$



Worked Example



Your notes



Your notes



a) Solve $\tan^2 x - 2 \tan x = 0$, in the interval $0^\circ \leq x \leq 360^\circ$.

Giving each solution correct to 1 decimal place.

b) i) Show that $11 \sin x = 5 \cos^2 x + 7$ can be written in the form $5 \sin^2 x + 11 \sin x - 12 = 0$.

ii) Find all the solutions of the equation

$11 \sin x = 5 \cos^2 x + 7$, in the interval $0^\circ \leq x \leq 360^\circ$.

Giving each solution correct to 1 decimal place.

a)

$$\tan^2 x - 2 \tan x = 0$$

$$\tan x (\tan x - 2) = 0$$

← FACTORISE

TWO POSSIBLE SOLUTIONS

$$\tan x = 0$$

$$\tan^{-1}(0) = 0$$

$$\tan x - 2 = 0$$

$$\tan x = 2$$

$$\tan^{-1}(2) = 63.43$$

$$63.4 + 180 = 243.4$$

Tan REPEATS EVERY 180°
SO ADD 180° TO FIND ALL SOLUTIONS

$x = 0, 180, 360$ AND $x = 63.4, 243.4$

b) i)

$$11 \sin x = 5 \cos^2 x + 7$$

$$11 \sin x = 5(1 - \sin^2 x) + 7$$

$$11 \sin x = 5 - 5 \sin^2 x + 7$$

$$5 \sin^2 x + 11 \sin x - 12 = 0$$

← USE $\cos^2 x = 1 - \sin^2 x$ TO GET EVERYTHING IN TERMS OF $\sin x$

← EXPAND THEN REARRANGE

← USING

ii) $5\sin^2 x + 11\sin x - 12 = 0$
 $(5\sin x - 4)(\sin x + 3) = 0$

FACTORISE USING $\sin x = s$ TO MAKE IT EASIER
 $(5s - 4)(s + 3)$

$5\sin x - 4 = 0$
 $5\sin x = 4$
 $\sin x = \frac{4}{5}$

$\sin x + 3 = 0$
 $\sin x = -3$

$11\sin x = 5\cos^2 x + 7$
 WRITTEN IN THE FORM
 $5\sin^2 x + 11\sin x - 12 = 0$
 FROM PART i)

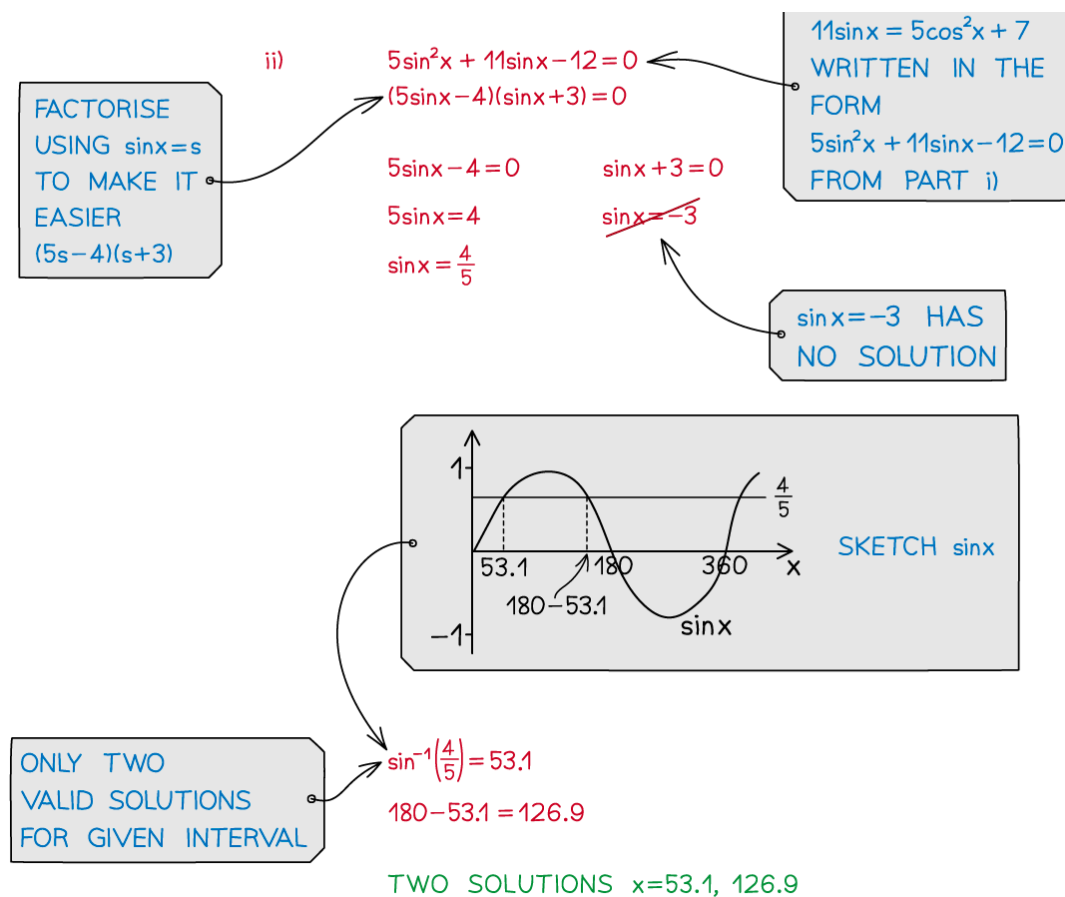
$\sin x = -3$ HAS NO SOLUTION

SKETCH $\sin x$

ONLY TWO VALID SOLUTIONS FOR GIVEN INTERVAL

$\sin^{-1}\left(\frac{4}{5}\right) = 53.1$
 $180 - 53.1 = 126.9$

TWO SOLUTIONS $x = 53.1, 126.9$



Strategy for Further Trig Equations

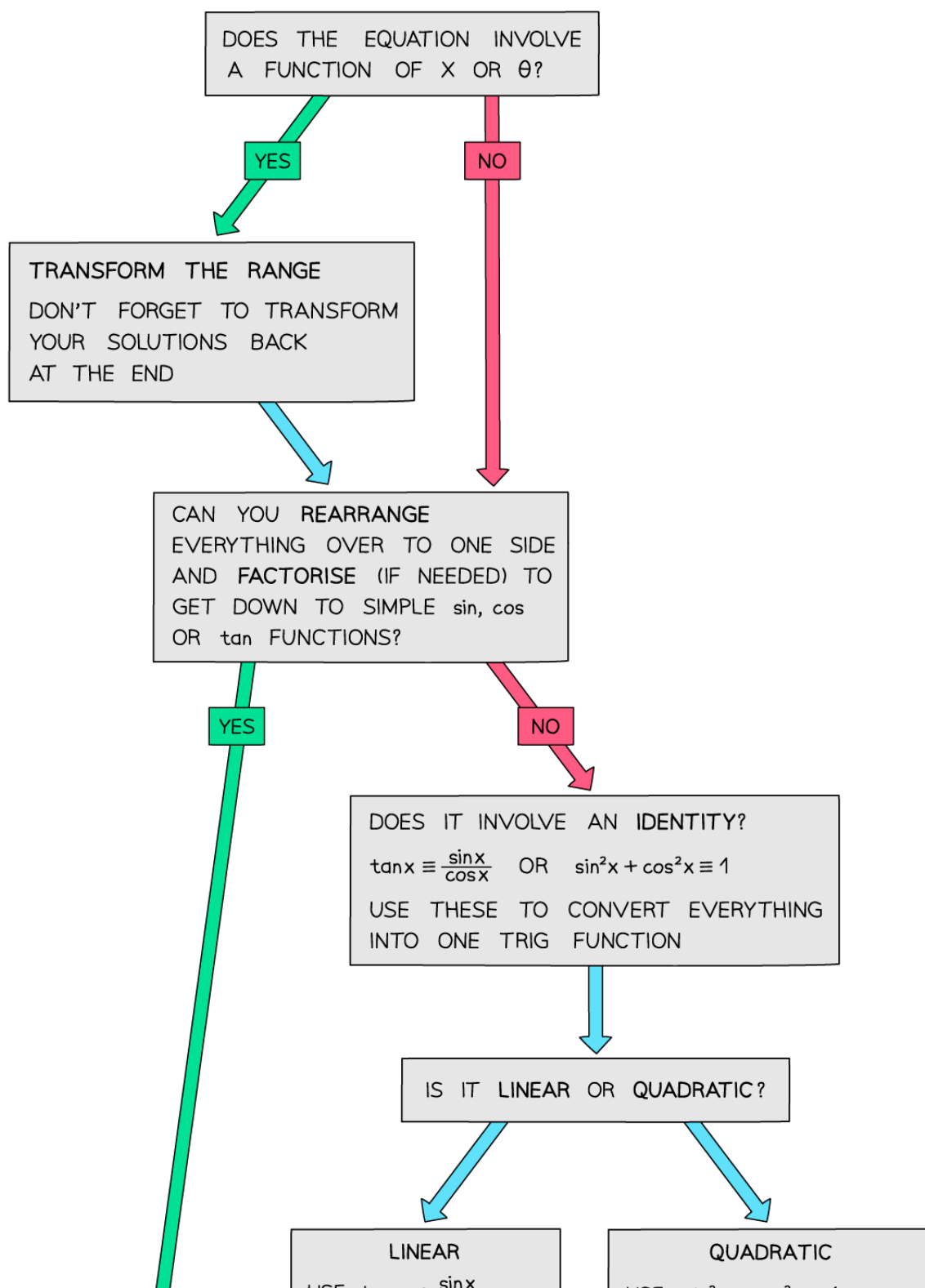
How to approach solving trig equations

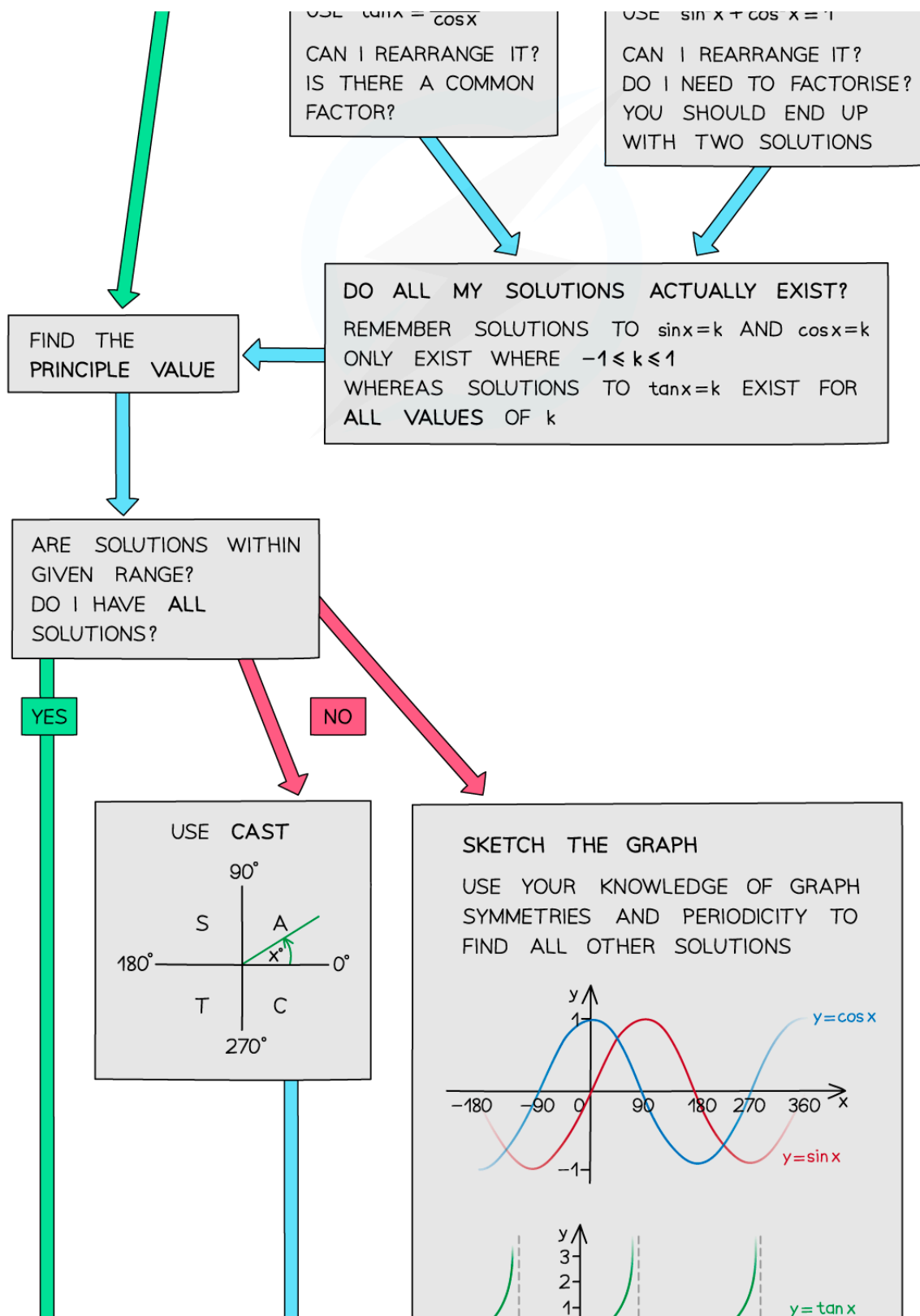
- You can solve trig equations in a variety of different ways
 - Sketching a graph (see **Graphs of Trig Functions**)
 - Using **trigonometric identities** (see **Simple Trig Identities** and **Further Trig identities**)
 - Using the **CAST** diagram (see **Linear Trig Equations**)
 - Factorising **quadratic** trig equations (see **Quadratic Trig Equations**)
- You may be asked to use **degrees or radians** to solve trigonometric equations
 - Make sure your calculator is in the **correct mode**

- Remember common angles
 - 90° is $\frac{1}{2}\pi$ radians
 - 180° is π radians
 - 270° is $3\pi/2$ radians
 - 360° is 2π radians
- If you're having trouble solving a trig equation, this flowchart might help:



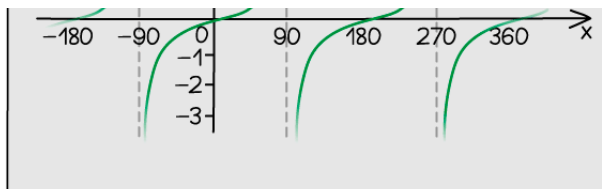
Your notes







Your notes



CHECK WHETHER YOU HAVE FOUND ALL SOLUTIONS AND THAT THEY ARE WITHIN THE GIVEN RANGE
(DON'T FORGET TO TRANSFORM YOUR SOLUTIONS BACK AT THE END IF THEY WERE FROM A TRANSFORMED RANGE)

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Examiner Tips and Tricks

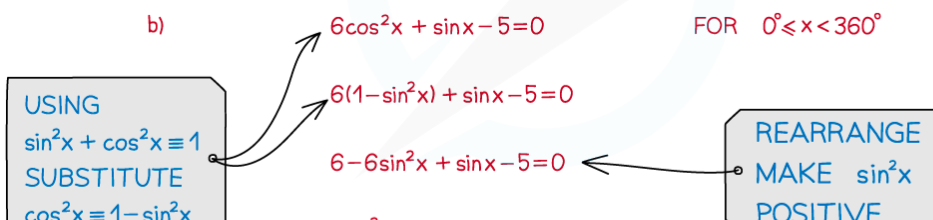
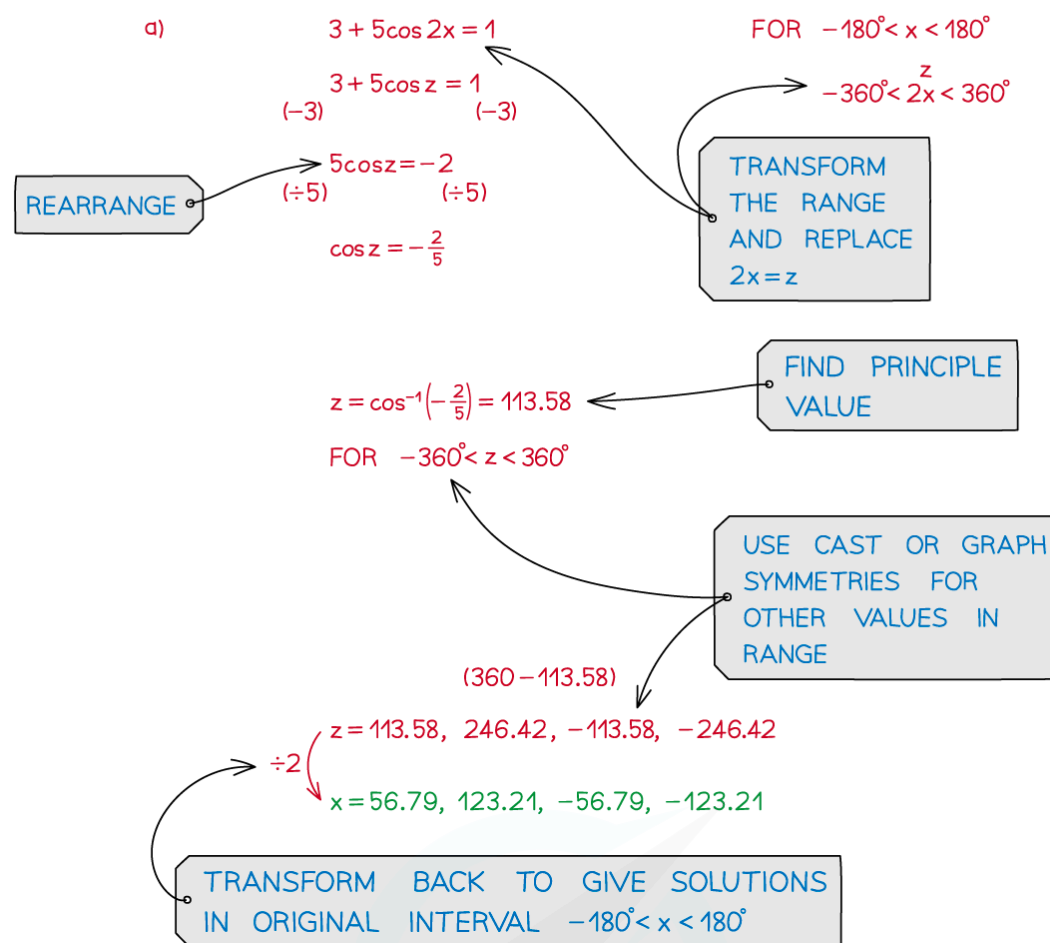
- Don't forget to check the function range and ensure you have included all possible solutions
- If the question involves a function of x or θ , make sure you transform the range first (and ensure you transform your solutions back again at the end!)



Worked Example

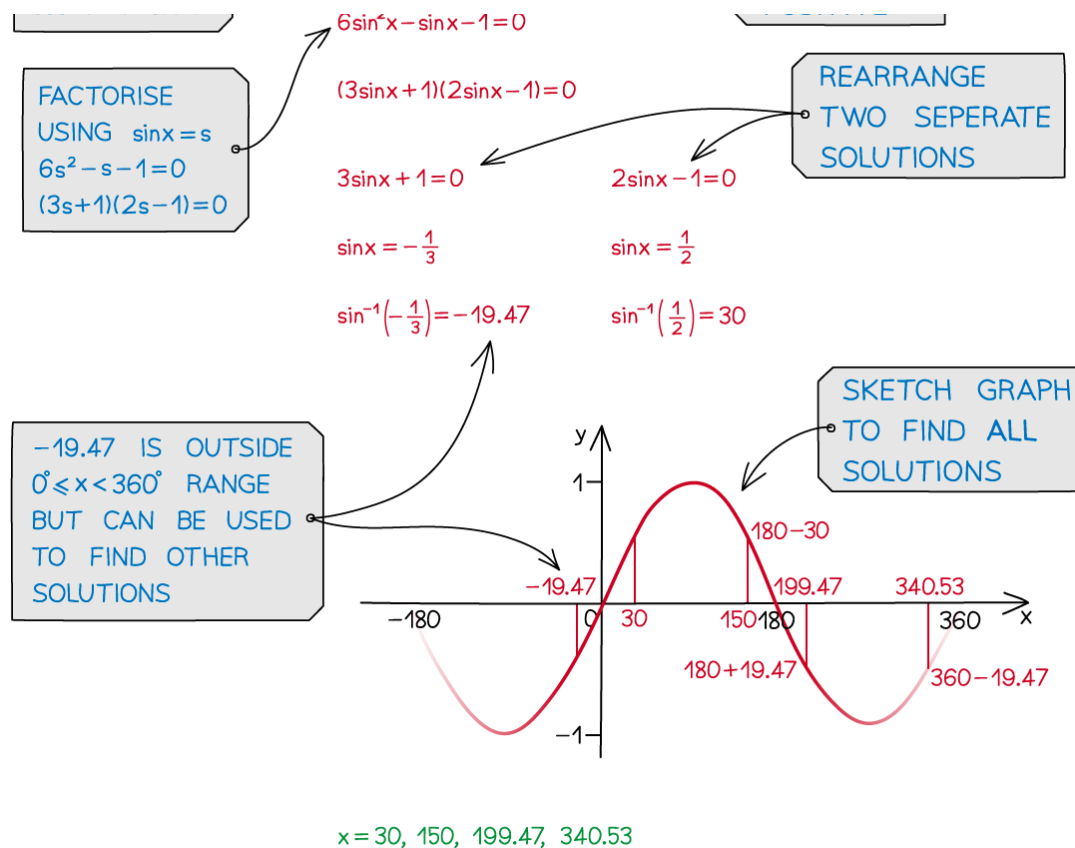


- a) Solve $3 + 5 \cos 2x = 1$, for $-180^\circ < x < 180^\circ$.
b) Solve $6 \cos^2 x + \sin x - 5 = 0$, for $0 \leq x < 360^\circ$.





Your notes



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Your notes

Trigonometric Proof

Trigonometric Proof

Proving trigonometric identities

- You can use trigonometric identities you already know to prove new identities
- Make sure you know the **simple trigonometric identities** and **further trigonometric identities**
 - $\sin^2 \theta + \cos^2 \theta = 1$
 - $\frac{\sin \theta}{\cos \theta} = \tan \theta$
 - $1 + \tan^2 \theta = \sec^2 \theta$
 - $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$
- To prove an identity start on one side and proceed step by step until you get to the other side
 - e.g. to prove that $(\sin \theta + \cos \theta)^2 = 1 + 2\sin \theta \cos \theta$
 - Start with the left hand side, and expand
 - $\sin^2 \theta + \cos^2 \theta + 2\sin \theta \cos \theta$
 - Simplify using the identity $\sin^2 \theta + \cos^2 \theta = 1$
 - $1 + 2\sin \theta \cos \theta$
 - This is now equal to the expression on the right, so the statement has been proven
- Make sure you are confident handling fractions and fractions-within-fractions



Your notes

e.g. SHOW THAT $\frac{1 + \tan \theta}{1 + \cot \theta} \equiv \tan \theta$

$$\begin{aligned} \frac{1 + \tan \theta}{1 + \cot \theta} &\equiv \frac{1 + \tan \theta}{1 + \frac{1}{\tan \theta}} \quad \leftarrow \cot \theta \equiv \frac{\cos \theta}{\sin \theta} \equiv \frac{1}{\tan \theta} \\ &\equiv \frac{1 + \tan \theta}{\left(\frac{\tan \theta + 1}{\tan \theta} \right)} \\ &\equiv (1 + \tan \theta) \times \frac{\tan \theta}{\tan \theta + 1} \\ &\equiv \tan \theta \end{aligned}$$

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- Always keep an eye on the 'target' expression – this can help suggest what identities to use



Examiner Tips and Tricks

- Don't forget that you can start a proof from either side – sometimes it might be easier to start with a particular side
 - If you get stuck – try starting from the other side instead!



Worked Example

Prove that:

$$\sec^2 x (\cot^2 x - \cos^2 x) = \cot^2 x$$

Use the definitions of $\sec x = \frac{1}{\cos x}$ and $\cot x = \frac{1}{\tan x} = \frac{\cos x}{\sin x}$, to rewrite the left hand side



Your notes

$$\frac{1}{\cos^2 x} \left(\frac{\cos^2 x}{\sin^2 x} - \cos^2 x \right)$$

Expand the brackets, using the fact that multiplying by $\frac{1}{\cos^2 x}$ is the same as dividing by $\cos^2 x$

$$\frac{1}{\sin^2 x} - 1$$

Use the definition of $\operatorname{cosec} x = \frac{1}{\sin x}$ to rewrite the expression

$$\operatorname{cosec}^2 x - 1$$

Use the identity $1 + \cot^2 x = \operatorname{cosec}^2 x$ to rewrite the expression

$$1 + \cot^2 x - 1$$

Simplify to achieve the right hand side

$$\cot^2 x$$